Pietro Freni

Studi

2020-2024 **PhD in Pure Mathematics**, *University of Leeds*
Relatori Vincenzo Mantova and H. Dugald Macpherson
Tesi Structural Investigations in some Classes of o-minimal Fields. https://etheses.whiterose.ac.uk/id/eprint/36168

2017-2019 **Laurea Magistrale in Matematica (M.S.)**, *Università di Pisa*, 110/110 cum laude
Relatore Alessandro Berarducci
Tesi Surreal Numbers, Transseries and Omega Maps.

2012-2016 **Laurea Triennale in Matematica (B.S.)**, *Università di Pisa*, 110/110 cum laude
Relatore Enrico Sbarra
Tesi Risoluzioni libere di ideali monomiali.

2007-2012 **Maturità Scientifica**, *Liceo Scientifico Archimede, Messina*, 100/100 L

Esperienza Professionale

2026-ora **Post-doc**, *Istituto di Matematica, Accademia delle Scienze Ceca*, Matematica pura

2025 **Ricercatore**, *Università di Leeds*, Matematica pura

2025 **Supplente**, *IISS Antonio Pesenti, Cascina (PI), Italy*, Matematica e Fisica

2020-2023 **Tutor di Matematica**, *Stem-up*

2020 **Supplente**, *ITCG Fermi Pontedera*, Informatica

Interessi di Ricerca

Teoria dei Modelli, o-minimalità, teoria delle categorie, logica matematica.

Pubblicazioni

2021 Berarducci, A., Freni, P. (2021). *On the value group of the transseries.* Pacific Journal of Mathematics, 312(2), 335-354. https://msp.org/pjm/2021/312-2/p04.xhtml, arxiv version: https://arxiv.org/abs/2101.04002

Preprint

2023 *On Vector Spaces with Formal Infinite Sums.* https://arxiv.org/abs/2303.08000, accepted for pubblication in Applied Categorical structures.

2024 *T-convexity, Weakly Immediate Types and T-λ-Spherical Completions of o-minimal Structures.* https://arxiv.org/abs/2404.07646

2024 *On the structure of T-λ-spherical completions of o-minimal fields.* https://arxiv.org/abs/2407.07442

2024 *T-convexly valued o-minimal fields are definably spherically complete.* https://arxiv.org/abs/2411.16706

2025 (j.w. Angus Matthews) *Residually Constructible Extensions.* https://arxiv.org/abs/2501.10033

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2025 *Absorbed Types and Derivations in Exponential o-Minimal Theories.* `https://arxiv.org/abs/2511.13447`

Poster

2022 On formal infinite linear combinations (Presented in different versions at: Model Theory and Applications, Cetraro 2022; Unimod 2022, Leeds).

Seminari

2025 *Cuts and Derivations in Exponential o-minimal Fields*. Leeds, Models Seminar, and Imperial College London, SEEMOD2025; *Expansions of Hahn fields by Algebras of generalized power series*. Napoli, INdAM intensive period on Model theory and tame expansions of topological fields, and Bologna, Logic, Categories, and Applications Seminar; *Strong vector spaces and some torsion theories of* $\mathrm{Ind}(\mathrm{Vect}^{\mathrm{op}})$. Bologna, Logic, Categories, and Applications Seminar. *Elementary (bounded) spherical completions of exponential o-minimal fields, and their structure.* Pisa, Logic seminar.

2024 *Weakly immediate types and spherical completions of o-minimal structures.* Burnley, 30th LYMoTS meeting; *T-convexity and T-λ-spherical completions of o-minimal structures.* Manchester Logic Seminar.

2023 *What Should Strong Vector Spaces be?* Online, ItaCa Fest May 2024, Leeds, Impromptu Leeds-MacMaster Model Theory days; *Summability in Algebras of Generalized Power Series.* Leeds, Postgraduate Logic Seminar; *How do you solve a system of polynomial equations?* Leeds, Postgraduate Pure Math Seminar.

2022 *What are Transseries?* Leeds, Postgraduate Pure Math Seminar; *Strong Vector Spaces* Minitalk at LYMOTS/SEEMOD meeting.

2021 *Around exponentiation in fields of generalized power series.* Leeds, Postgraduate Logic Seminar; *Gabrielov's theorem and the model completeness of reals with restricted analytic functions.* Leeds, Postgraduate Pure Math Seminar.

Esperienza di Insegnamento

2021 Università di Leeds. Tutorato: Numbers and Systems.

2022 Università di Leeds. Tutorato: Sets, Sequences and Series. Esercitazioni: Further Linear Algebra.

2023 Università di Leeds. Tutorato: Sets, Sequences and Series.

Altre esperienze

2025 Orginizzatore del the Models Seminar all'Università di Leeds.

2023 Organizzatore della British Postgraduate Model Theory Conference 2023.

2022-2023 Organizzatore del Leeds Postgraduate Logic Seminar.

2022 Aiuto nell'organizzazione del programma Unimod 2022.

Borse e Premi

2020-2024 School of Mathematics Full-Time Doctoral Scholarship at the School of Mathematics, University of Leeds.

2010-2012 Bronze Medal at the Italian Mathematics Olympiads in years 2010, 2011, 2012.

Abilità informatiche

Coding C, C#, octave, SQL (basic experience with C++, Java, javascript, M2, AGDA).

Other LaTeX, html, css, Excel.

Lingue

Fluente	Italiano (madrelingua), Inglese.
Buono	Tedesco.